INTERNATIONAL CONFERENCE ON INNOVATIVE PATHWAYS TO SUSTAINABLE BUSINESS: AI, ESG & RESPONSIBLE LEADERSHIP

EduMatrix: Institutional Academic and Placement Intelligence System

Adithya C S[1] | Manikanta[1] | Pushanth S B[1] | **Rithesh N D[1]* (Presenter)** | Nidhishree M P[2]

[1]Department of Artificial Intelligence and Machine Learning

[2]Assistant Professor, Department of Artificial Intelligence and Machine Learning

Vivekananda College of Engineering and Technology, Puttur, Karnataka, India

Hosted by: Srinivas Institute of Technology, Valachil, Mangaluru • Date: 15 October 2026

CONTEXT & PROBLEM FORMULATION

The Institutional Communication & Placement Dilemma

1. Fragmented Communication

Decentralized Channels & Missed Deadlines

- Academic circulars, internal exam dates, and placement notices are scattered across WhatsApp, emails, and physical boards.
- Leads to severe communication latency, information loss, student confusion, and missed critical application deadlines.
- Students lack a centralized dashboard for tracked deadlines.

2. Faculty Exam Scheduling

Manual Invigilation & Timetable Conflicts

- Heads of Departments (HODs) manually assign examination invigilation duties from raw schedule sheets.
- High risk of timetable scheduling collisions, room overlap, and unequal faculty duty distributions.
- No automated conflict detection or instantaneous duty notification mechanism for faculty members.

3. Manual Placement Screening

Unstructured Resumes & Skill Blindspots

- Placement cells manually screen hundreds of unstructured PDF resumes against company criteria.
- Time-intensive, subjective evaluation without quantitative ATS scoring or benchmarked skill compatibility.
- Students receive zero diagnostic feedback on missing technical proficiencies before recruitment drives.

Research Gap & Core Motivation: Existing solutions function in isolated silos (messaging apps vs. ERPs vs. job portals). Higher education institutions lack an integrated, intelligent platform that seamlessly connects real-time communication with automated document intelligence and predictive placement screening.

SYSTEM SCOPE & KEY DELIVERABLES

Core Research Objectives (Supported by Paper)

Objective 1: Timetable & Circular Processing
Automatically extract examination dates, subjects, timings, and critical deadlines from uploaded PDF/image circulars and timetables using OCR and computer vision text processing.

Vision Transformer & OCR

Objective 2: Faculty Examination Duty Management
Assist HODs in assigning faculty invigilation duties, detecting schedule clashes across halls/dates, and automatically dispatching digital duty allocations to faculty.

Automated Conflict Resolution

Objective 3: Resume ATS Scoring & Skill-Gap Analysis
Parse uploaded student resumes, extract technical skills/projects, evaluate profiles against company placement requirements using vector similarity, and provide ATS scores with skill-gap feedback.

LayoutLMv3 & Cosine Similarity

Objective 4: Real-Time Communication & Reminders
Deliver instant in-app alerts, categorized deadline notifications (upcoming, due today, missed), and direct student-faculty-HOD messaging using persistent WebSocket channels.

WebSocket & Redis Streams

Objective 5: Role-Based Institutional Dashboards
Provide dedicated, secure web interfaces for Students, Faculty, HODs, Placement Officers, and Administrators with strict JWT-enforced Role-Based Access Control (RBAC).

Secure RBAC & Multi-Portal UX

SYSTEM FRAMEWORK & TECHNICAL PIPELINE

EduMatrix Proposed Architecture & Methodology

Stage 1: Interface & Ingestion
Heterogeneous Data Collection

- Multi-Role Frontend: React, Vite, Tailwind CSS with JWT auth.
- Academic Ingestion: Unstructured PDF/image circulars & exam timetables.
- Placement Ingestion: Student PDF resumes & recruiter job descriptions.
- Roster Data: Faculty departmental lists and course schedules.

Stage 2: Vision & OCR Pipeline
Document Preprocessing

- Image Preprocessing: OpenCV grayscale, adaptive thresholding, denoising.
- OCR Engine: Tesseract OCR extracts raw unstructured textual content.
- Regex & Parsing: Identifies course codes, session timings, and deadlines.
- Cleaned metadata queued in PostgreSQL.

Stage 3: Machine Learning Engine
AI Models & Analytics

- Table Transformer: Vision Transformer detects table boundaries & duty slots.
- LayoutLMv3 + spaCy: Multimodal resume parsing & technical skill extraction.
- Cosine Similarity ATS Scoring: Compares candidate vector r against job vector j.
- BERT & DL Classifiers: Student intent classification & sentiment analysis.

Stage 4: Real-Time Services
WebSocket & Role Dashboards

- Real-Time Delivery: WebSockets & Redis for sub-second alert broadcasting.
- Categorized Alerts: Upcoming, Due Today, and Missed academic deadlines.
- Duty Allocation: Clash-free invigilation duty roster dispatched to faculty.
- Placement Module: ATS scoring and skill-gap recommendations.

Core ATS Match Formulation (Section II-B): $ATS\ Score = [\ (\ r \cdot j\ )\ /\ (\ ||r||\ ||j||\ +\ \varepsilon\ )\ ]\ \times\ 100$

Where r represents the extracted student resume feature vector, j represents the target company job requirement vector, and ε prevents division by zero to guarantee stable, objective scoring.

EMPIRICAL EVALUATION & EXPERIMENTAL VALIDATION

Key Findings: Model Accuracy & Operational Benchmarks

Table 2: Machine Learning Model Comparison (from Paper)

Parameter	Table Transformer	Resume Parser	Intent Classifier	Sentiment Classifier
Model Type	Vision Transformer	Document NLP / Multimodal	Fine-tuned BERT	Deep Learning NLP
Application	Timetable / Table Extraction	Resume Parsing & ATS Scoring	Student Query Classification	Feedback Analysis
Training Acc.	96.50%	96.20%	97.80%	95.40%
Testing Acc.	**94.20%**	**93.90%**	**95.60%**	**93.10%**

Key Experimental Observations

✓ 94.20% Table Detection Accuracy
Vision Transformer successfully isolates table cells from complex academic circular layouts, eliminating schedule collisions during invigilation duty assignment.

✓ 93.90% Resume & ATS Precision
Multimodal extraction detects hard and soft competencies, reliably scoring candidate suitability against recruiter criteria without manual human bias.

✓ 95.60% Automated Query Resolution
Fine-tuned BERT intent model routes institutional inquiries (exams, marks, eligibility) accurately, reducing administrative support ticket load.

✓ Zero-Latency Real-Time Alerts
WebSockets deliver instantaneous notifications for deadline statuses (upcoming, due today, missed), ensuring 100% awareness across student cohorts.

CONFERENCE THEME: AI, ESG & RESPONSIBLE LEADERSHIP

Practical, Organizational & ESG Implications

Environmental Sustainability (E in ESG)
Paperless Campus Governance & Zero Physical Drag

- Automated PDF/image circular ingestion eliminates physical paper printing and manual board postings across departments.
- Digital deadline tracking and mobile/web notices minimize institutional carbon footprint and physical administrative overhead.

Social Responsibility & Equity (S in ESG)
Faculty Well-Being & Unbiased Career Mobility

- Algorithmic invigilation assignment ensures balanced, conflict-free workload distribution, preventing faculty burnout.
- Objective ATS resume scoring democratizes placement access, eliminating subjective hiring bias and exposing transparent skill gaps.

Responsible Governance (G in ESG)
Transparent, Auditable Institutional Operations

- Strict Role-Based Access Control (RBAC) across 5 distinct portals ensures data segregation, security, and full accountability.
- Placement coordinators gain real-time visibility into student readiness metrics, enabling targeted institutional training initiatives.

Implementation Feasibility & Limitations
Production Deployment on Standard Campus Infrastructure

- Demonstrated on standard hardware (Intel Core i5, 8–16 GB RAM, PostgreSQL) without requiring expensive enterprise clusters.
- Current Scope: Evaluated on institutional English document templates; future deployment requires multilingual regional circular parsing.

SUMMARY & FUTURE HORIZONS

Conclusion and Research Directions

Key Research Takeaways

✓ Unified Institutional Platform: EduMatrix bridges the critical disconnect between academic administration, faculty duty allocation, and student career readiness.

✓ High-Precision Document Intelligence: Demonstrated 94.20% accuracy in timetable extraction and 93.90% accuracy in automated ATS resume scoring.

✓ Operational Efficiency: Replaced fragmented WhatsApp/email channels with real-time WebSocket alerts, eliminating missed deadlines and invigilation clashes.

✓ Democratized Career Guidance: Empowers students with actionable skill-gap diagnostics and objective company compatibility metrics.

Future Research Directions

➔ Multilingual Document Extraction: Integrating Indic NLP models to process state government and regional university circulars.

➔ Cross-Campus Institutional Federation: Expanding the architecture to support multi-college university clusters and centralized placement pools.

➔ AI Mock Interview & Voice Evaluation: Incorporating conversational speech-to-text models to simulate real-time technical interviews and evaluate soft skills.

➔ Predictive Placement Forecasting: Machine learning models to track longitudinal student academic trajectories and predict multi-tier hiring outcomes.

Thank You for Your Time and Attention!

We warmly invite Questions, Feedback, and Discussion.

Presenting Author: Rithesh N D • VCET Puttur • Email: pushanthsb@gmail.com